

# Complete guide on home spirulina farming



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## **Words from the author**

This manual is intended for those who pursue growing spirulina at home due to their love for science, their will to embrace new sustainable eating practices or their unconditional love for spirulina. In this guide you will find everything you should know to start producing spirulina at home successfully. By following the easy-steps mentioned in this manual you will be able to enjoy your fresh spirulina regularly.

My name is Alejandro Espi and I am the creator of this guide. I am a holder of a BSc with Honours in Biological Sciences, an MSc in Clinical Microbiology and an MSc in Aquaculture, Environment and Society. Throughout my education and professional career I have always aimed at solving the humanitarian and environmental challenges that our societies are facing. This interest made me realise of the potential of certain microorganisms as a food source due to their nutritional characteristics and the low environmental impacts associated with their production.

If you would like to receive any consultancy regarding how to grow spirulina please, feel free to send me an email to:

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## Donations

All the information provided in this manual is the product of countless hours of reading and experimentation. If you want to support this investigation and future ones please, send your donations to:

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**Dogecoin:**

DDPrHukBhUbUepZQDXSKJeKBqGKBAXDjEB



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Those who inspired others to protect the environment and people in vulnerable situations, thank you. Those who work countless hours for the sake of science, thank you.

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# Vocabulary

**Cultivation tank:** another term to refer to a growing vessel i.e. the container used to grow spirulina.

**Culture medium:** the nutrient-rich medium in which microorganisms are grown. It can be considered as the food of the spirulina.

**Growing vessel:** another term to refer to a cultivation vessel i.e. the container used to grow spirulina.

**Photobioreactor:** this is just a fancy word that refers to a cultivation tank in which light is used to grow organisms.

**Secchi disk:** this is a disk attached to a ruler that allows you to measure the turbidity of a liquid culture.

**Turbidity:** measure of the cloudiness of the culture. This parameter can be used as an indicator for the amount of spirulina in the culture.

**Zarrouk's medium:** the culture medium of choice to grow spirulina.

# 1.Introduction

## 1.1 What is spirulina?

*Arthrospira platensis* and *Arthrospira maxima*, aka spirulina, are two species of cyanobacteria (blue-green algae) (Tomaselli, 2002) that are used as a nutritional supplement due to their nutritional profiles. In this respect, a high economic potential has been attributed to these microalgae due to their high content in proteins (50–70%), lipids (13%) and pigments (phycocyanin and chlorophyll) (Ferreira *et al.*, 2010).

These two species of blue-green algae reproduce by dividing into two along a single plane, a process called binary fission. Spirulina forms spirals known as trichomes which have an anti-clock wise shape (**Fig. 1**). These spirals float freely in their environment and have a length and a width of 12-72  $\mu\text{m}$  and 30-70  $\mu\text{m}$  respectively (Tomaselli, 2002).



**Figure 1.** Micrograph of a spirulina trichome taken with light microscope.

Spirulina can be found growing naturally in alkaline and warm waters of Asia, Africa, Europe and America, where it is able to thrive in both fresh and seawater (Gutiérrez-Salmeán *et al.*, 2015).

## 1.2 Traditional uses of spirulina

Spirulina used to grow in large quantities at the Texcoco lake, Mexico (Gutiérrez-Salmeán *et al.*, 2015). There, it was consumed as a sort of dry cake (Abdulqader *et al.*, 2000) called *tecuitlatl* during the prehispanic era (Usharani *et al.*, 2012) by the Aztecs about 1300AD (Abdulqader *et al.*, 2000). Later on, during the times of conquest, it was harvested, processed and sold for human consumption. Regretfully, the use of spirulina as food in Mexico disappeared over time (Gutiérrez-Salmeán *et al.*, 2015).

Spirulina has also been historically consumed by the Kanembu tribes of Chad (European Commission, 2020) and Niger (Abdulqader *et al.*, 2000) that inhabit the areas surrounding the lakes Chad and Kossorom. There, spirulina is used to prepare a sort of cake called *dihé* (Fig. 2) that is later used to prepare a vegetable or fish broth that is referred to as *la souce*. These broth is consumed with staple cereals (Abdulqader *et al.*, 2000).



**Figure 2.** Dried spirulina in the form of *dihé* (Abdulqader *et al.*, 2000).

## 2. Nutritional profile of spirulina

Spirulina has a rich-nutritional value in terms of micro and macro-nutrients. These include, among others, proteins of high quality, iron, gamma-linoleic acid, vitamins, minerals, sulphated polysaccharides and pigments such as phycocyanin. These nutritional characteristics have been used to point at spirulina as a potential functional food i.e. a food that can help specific body functions by promoting health or reducing the likelihood of disease. However it is important to note that the nutritional profile of spirulina varies according to its cultivation method (Gutiérrez-Salmeán *et al.*, 2015).

The **table 1** displays the nutritional profile of spirulina in terms of micro and macronutrients as stated in a peer-review paper written by Gutiérrez-Salmeán *et al.*, (2015). In this article it was stated that the nutritional analyses of spirulina were performed by an independent laboratory and Earthrise Nutritionals LLC (CA, U.S.A).

### 2.1 Protein

The protein content of spirulina is exceptional since it makes about 60-70% of its total dry mass (mass with no water). This is above and beyond the protein content of many other plant-based food sources in which it normally makes about 35% of their total dry mass (Gutiérrez-Salmeán *et al.*, 2015).

The protein present in spirulina is also of high quality since almost half of it comprises essential amino acids. This makes these microalgae a good source of complete protein. Finally, it is important to mention that the assimilation of the protein present in spirulina is much higher than that of other plant sources (Gutiérrez-Salmeán *et al.*, 2015), making it a good source of high quality protein that can be readily assimilated.

### 2.2 Lipids

About 5-10% of the dry weight of spirulina are lipids. However it is important to note that most of these lipids are regarded as essential fats to humans which include gamma-linolenic, linoleic and oleic acids.

Several peer-reviewed articles have stated that spirulina is the vegetable source with the highest amount of gamma-linolenic acid, a lipid required to produce mediators of inflammatory and immune responses (Gutiérrez-Salmeán *et al.*, 2015).

### 2.3 Vitamins

*Spirulina is a good source of vitamins and carotenoids* (pigments present in algae that have lipophylic antioxidative properties). Specifically, spirulina has been regarded as a great

source of beta-carotene (700-1700 mg/kg) which upon being absorbed is converted into vitamin A (Gutiérrez-Salmeán *et al.*, 2015).

## 2.4 Minerals

*Spirulina* presents a number of different minerals including iron, calcium and phosphorous. Specifically, spirulina is regarded as a good iron source with respect to other vegetable sources since in addition to its iron levels, it does not present compounds that reduce its absorption (Gutiérrez-Salmeán *et al.*, 2015).

The calcium and phosphorous levels of spirulina are similar to those found in milk, being present in ratio (Ca:P) compatible with the maintenance of bone health. In addition, as in the case of iron, the calcium present in spirulina has a high availability since these microalgae do not present compounds that lower its absorption (Gutiérrez-Salmeán *et al.*, 2015).

**Table 1.** Nutritional characteristics of spirulina powder (for every 100 grams) as shown by Gutiérrez-Salmeán *et al.*, (2015).

|                                       |       |                                               |            |
|---------------------------------------|-------|-----------------------------------------------|------------|
| <b>Macronutrients</b>                 |       | <b>Vitamins</b>                               |            |
| Calories                              | 373   | Vitamin A (as $\beta$ -carotene) <sup>b</sup> | 352.000 IU |
| Total fat (g)                         | 4.3   | Vitamin K                                     | 1090 mcg   |
| Saturated fat                         | 1.95  | Thiamine HCL (Vitamin B1)                     | 0.5 mg     |
| Polyunsaturated fat                   | 1.93  | Riboflavin (Vitamin B2)                       | 4.53 mg    |
| Monounsaturated fat                   | 0.26  | Niacin (Vitamin B3)                           | 14.9 mg    |
| Cholesterol                           | < 0.1 | Vitamin B6 (Pyridox. HCL)                     | 0.96 mg    |
| Total carbohydrate (g)                | 17.8  | Vitamin B12                                   | 162 mcg    |
| Dietary fiber                         | 7.7   | <b>Minerals</b>                               |            |
| Sugars                                | 1.3   | Calcium                                       | 468 mg     |
| Lactose                               | < 0.1 | Iron                                          | 87.4 mg    |
| Protein B                             | 63    | Phosphorus                                    | 961 mg     |
| <b>Essential amino acids (mg)</b>     |       | Iodine                                        | 142 mcg    |
| Histidine                             | 1000  | Magnesium                                     | 319 mg     |
| Isoleucine                            | 3500  | Zinc                                          | 1.45 mg    |
| Leucine                               | 5380  | Selenium                                      | 25.5 mcg   |
| Lysine                                | 2960  | Copper                                        | 0.47 mg    |
| Methionine                            | 1170  | Manganese                                     | 3.26 mg    |
| Phenylalanine                         | 2750  | Chromium                                      | <400 mcg   |
| Threonine                             | 2860  | Potassium                                     | 1,660 mg   |
| Tryptophan                            | 1090  | Sodium                                        | 641 mg     |
| Valine                                | 3940  | <b>Phytonutrients</b>                         |            |
| <b>Non-essential amino acids (mg)</b> |       | Phycocyanin (mean) <sup>b</sup>               | 17.2%      |
| Alanine                               | 4590  | Chlorophyll (mean) <sup>b</sup>               | 1.2%       |
| Arginine                              | 4310  | Superoxide dismutase (SOD)                    | 531,000 IU |
| Aspartic acid                         | 5990  | Gamma linolenic acid (GLA)                    | 1080 mg    |
| Cystine                               | 590   | Total carotenoids (mean) <sup>b</sup>         | 504 mg     |
| Glutamic acid                         | 9130  | $\beta$ -carotene (mean) <sup>b</sup>         | 211 mg     |
| Glycine                               | 3130  | Zeaxanthin                                    | 101 mg     |
| Proline                               | 2380  |                                               |            |
| Serine                                | 2760  |                                               |            |
| Tyrosine                              | 2500  |                                               |            |

<sup>a</sup> Most data are based on recent analysis by third-party laboratory.

<sup>b</sup> In Earthrise Nutritional LLC.

### 3. How to grow your own spirulina at home

#### 3.1 Materials

The term "photobioreactor" is just a fancy word that refers to a cultivation tank in which light is used to grow organisms. This is key for the successful cultivation of your spirulina since it is a photosynthetic organism (uses light to make its own food). In addition to light, temperature, aeration and pH are also parameters that should be controlled in your photobioreactor when growing your spirulina to make sure that your green spirals get everything they need. Now let's take a look at each of the components that you will need to build your reactor and ensure that your farming goes well.

**Transparent glass container:** This is where your spirulina will be cultivated. The container should be transparent, made of glass and as thin as possible in order to increase the light penetration into the culture (**Fig. 3a**). It is recommended that the container has a dispenser valve in order to facilitate the harvesting process.

**Artificial lighting:** Once your culture has reached the aimed turbidity (when the Secchi stick is not readable below its 5cm mark) it is advisable to place the tank under direct sunlight or to apply artificial lighting to ensure that the spirulina receive the light it needs. An adequate amount of light is crucial for the success of your culture since too little will slow down the cultivation process and too much may even cause the death of the culture. If you decide to use artificial lighting, the white light of a 12W LED lamp (**Fig.3b**) shone directly to the culture will work fine.

**Secchi stick:** This is just circular disk attached to a ruler. This instrument is used to calculate the turbidity of the spirulina culture since this parameter is an indicator for amount of spirulina in suspension. In order to check the turbidity of the culture, the Secchi disk (**Fig.3c**) should be immersed in the culture and the depth at which the disk is not longer visible recorded.

**Thermostat:** Numerous studies have shown that these microalgae can be successfully cultivated at temperatures at about 30-32 °C. Thus, it is recommended that a thermostat (**Fig.3d**) is used to maintain in the culture the optimal temperature for the growth of spirulina. You may be able to find this instrument at aquarium shops.

**Air pump with a silicon tube:** An adequate aeration is key for the success of any spirulina culture, especially if you decide to grow spirulina in a photobioreactor as described in this manual. In this respect, the air pump and a silicon tube (**Fig.3e**) are the instruments of choice to release bubbles into the culture. The bubbles produced by the pump are important for:

- Mixing the culture to ensure that all spirulina spirals are equally exposed to light.
- Preventing the spirulina from settling at the bottom of the tank.
- Impeding the formation of spirulina clumps (**Fig 4**).
- Maintaining the pH that the spirulina needs.
- Supplying some of the carbon dioxide/oxygen that is required for the spirulina to grow.

The aeration should just be strong enough to ensure that all your spirulina circulates around the growing vessel. An aeration that is too strong will damage the spirulina spirals and an aeration that is too weak will reduce the light penetration in the culture and lead to the formation of spirulina clumps.

You will be able to find this gadget in aquarium shops.



**Figure3.** Basic instruments required for spirulina production: transparent cultivation tank made of glass (a), LED lamp of white light with a power of 12 Watt (b), a Secchi stick (c), air pump (d) and a thermostat (e).



a



b

**Figure4.** Formation of spirulina clumps in the culture (shown as dark green lumps) due to inadequate mixing (a). Spirulina culture without apparent formation of clumps (b).

**Inorganic nutrients:** These inorganic substances will be used to prepare the food that the spirulina needs to thrive. As you may have heard, we are what we eat and spirulina is not an exception. Thus, we need to supply these microalgae with the substances it needs to boost their nutritional profile.

**Measuring scale:** This instrument will be key to prepare the " food " that you need to feed your spirulina culture. In this respect, you will need to measure the right amounts of the different nutrients to make sure that the culture medium is balanced. Since some of the nutrients are needed in very tiny amounts, it is recommended that the scale has an accuracy of 0.01 g (**Fig.5**).



**Figure 5.**Electronic balance with an accuracy of 0.01g as shown by the two decimal places.

**Filter cloth:** A filter cloth with a pore size of 30 microns (**Fig.6**) is required to successfully separate the culture medium from the spirulina during harvesting. In order to facilitate the filtration process it is recommended that the filtration mess is set over a porous tray before pouring the spirulina culture over it.



**Figure6.** Recommended instruments for an easy and successful filtration of the spirulina culture: a 30 microns filtration cloth (shown in white) and a porous tray (shown in green).

**Thermometer:** This tool is important for monitoring the temperature of the culture to ensure that your spirulina is cultured under the right conditions. An adequate temperature is crucial to prevent mortality in the culture and to boost its growth. You really do not want to mess up in this one.

**pH meter/pH testing paper strips:** A pH of  $9.5 \pm 0.5$  should be maintained in the culture throughout the farming process to boost the growth of the spirulina and to increase the resilience of the culture to potential invading microorganisms. You should monitor regularly the pH of your culture using an electronic pH meter or pH testing paper strips since this will give you an idea of the corrections you should implement. In the case that your culture has a pH lower than 9 you should add new culture medium and ensure that the adequate aeration is given.

### 3.2 Nutrients

As you may have heard, we are what we eat and spirulina is not an exception. If you want your spirulina to have a rich nutritional profile and to grow well you should feed it with a vast array of "high quality foods".

It is recommended growing the spirulina in what is known as Zarowk's medium. This medium will give to your spirulina everything it needs to thrive and also it will keep other potential invading microorganisms away. You can buy directly the powder of culture medium from different suppliers; nonetheless, it is much cheaper to prepare it by yourself.

In order to prepare 1 liter of culture medium you should dissolve the ingredients shown in the **table 2** into 1 liter of non-chlorinated water (either distilled water or drinking water).

**Table 2.** Recipe for the Zarrouk's medium. Amount of inorganic nutrients in grams that should be dissolved per litre of non-chlorinated water (Dineshkumar *et al.*, 2016).

| Inorganic nutrient                                                       | g/liter |
|--------------------------------------------------------------------------|---------|
| Sodium Bicarbonate (NaHCO <sub>3</sub> )                                 | 16.8    |
| Sodium Nitrate (NaNO <sub>3</sub> )                                      | 2.5     |
| Dipotassium Phosphate (K <sub>2</sub> HPO <sub>4</sub> )                 | 0.5     |
| Potassium Sulphate (K <sub>2</sub> SO <sub>4</sub> )                     | 1       |
| Sodium Chloride (NaCl)                                                   | 1       |
| Magnesium Sulphate Heptahydrate (MgSO <sub>4</sub> ·7H <sub>2</sub> O)   | 0.2     |
| Ferrous Sulphate (FeSO <sub>4</sub> )                                    | 0.01    |
| Tetrasodium Salt Of Ethylene Diamine Tetra Acide (EDTA·NO <sub>4</sub> ) | 0.08    |
| CaCl <sub>2</sub> · 2H <sub>2</sub> O                                    | 0.04    |

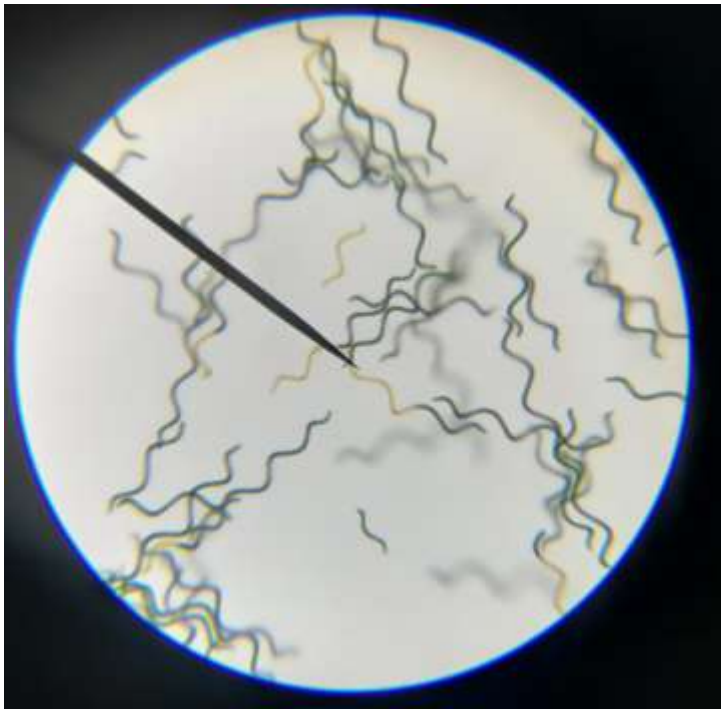
### 3.3 Farming process

Farming spirulina is a straight-forward process as long some basic rules and steps are followed. Think about them as a cooking recipe. If you want to make a cake, you cannot miss any step; otherwise, your final product may not have a good appearance, flavour or texture.

Before starting your spirulina culture, make sure that your culture vessel, the silicon tube of the air pump and the thermostat are well cleaned and rinsed with non-chlorinated water.

### 3.3.1 Culture conditions

The spirulina culture should be kept under specific conditions in order to speed up the cultivation time, to prevent contamination by other microorganisms and to reduce mortality in the culture. Should mortality occurs by any reason, the culture will turn yellow due to the decolouration of the spirulina spirals (**Fig 7**).



**Figure7.** Mortality-related discoloured appearance of spirulina spirals as seen under the light microscope.

The table below shows the optimal range values for the most important parameters that should be controlled in each cultivation tank.

**Table3.** Conditions required for the optimal growth of spirulina.

| Variable    | Value                                                                   | Tool to regulate it/check it                                                                          | Reference                          |
|-------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------|
| Light       | 120 micromol photons/m <sup>2</sup> s<br>i.e. 8.9 Klux<br>approximately | 40 W-<br>fluorescent<br>lamps<br><br>LED lights (12 W)                                                | (Ferreira <i>et al.</i> , 2010)    |
| Temperature | 30 ±2                                                                   | Thermostat                                                                                            | (Dineshkumar <i>et al.</i> , 2016) |
| pH          | 9.5 ±0.5                                                                | Daily addition of CO <sub>2</sub><br><br>Use of Zarrouk medium during cultivation due its alkalinity. | (Ferreira <i>et al.</i> , 2010)    |
| Air         | Weak aeration throughout the culture via air bubbles                    | Air pump                                                                                              | (Ferreira <i>et al.</i> , 2010)    |

### 3.3.2 Inoculation of your starter culture

1. Clean thoroughly with soap the cultivation vessel, the silicon tubing of the air pump and the thermostat. Then, rinse everything with non-chlorinated drinking water.
2. Set the silicon tubing of the air pump and the thermostat in the growing vessel where you plan to grow spirulina (**Fig.8a**).

**Important note:** In order to facilitate the adequate position of the silicon tube in the tank during the cultivation process, it is recommended to attach the tube to the thermostat using a clump (**Fig.8b**).



**Figure8.** Recommended setting of the silicon tube of the air pump and the thermostat in the cultivation tank (a). Silicon tube attached attached to a thermostat by a clump (b)

3. Prepare 1 litre of Zarrouk's medium for each litre of fresh spirulina culture that is in your possession. Look at the point **3.2** to find the recipe of this culture medium.
4. Pour at least half of prepared Zarrouk's culture medium into the cultivation tank.
5. Turn on the thermostat and the air pump.
6. Pour your fresh spirulina culture into the cultivation tank containing the culture medium.

7. Pour the remaining prepared culture medium into the cultivation tank until the Secchi disk is visible up to the 5cm reading. At this turbidity level (5 cm mark), your spirulina is sensitive to light. Thus, make sure that it is not exposed to direct sunlight during the peak hours or to very intense artificial lighting. To avoid this, at this stage it is recommended to keep your cultivation vessel in a place with no direct natural light. Nonetheless, feel free to apply artificial lighting (mentioned in **table 3**) to boost the photosynthetic activity of your spirulina so it can grow faster.

**Important note:** If you decide to use artificial lighting, the white light of 12W LED lamp shone directly to the culture will work fine. Feel free to choose the position of the LED lamp; however, you should remember that the higher the exposition of the spirulina to the light generated by the lamp, the faster it will grow. In the pictures presented in this manual it is possible to appreciate that the lamp is set just below the tank. This will make your design attractive in addition to efficient (**Fig 4**).

8. Monitor the turbidity of your initial culture on a daily basis with the Secchi stick. This will allow you to know when your culture is ready to increase its production.

**Important note:** Remember to keep also track of the pH of your culture to make sure everything is working properly.

### 3.3.3. Increasing your production

Your culture will be ready to increase its production once the Secchi disk is not visible at a depth of about 2-3cm.

1. Once your culture is ready for production, prepare and pour new culture medium into the growing vessel until the Secchi disk is visible up to a depth of 5cm. Look at the point **3.2** to find the recipe of this culture medium.

2. Repeat the previous step every time that the Secchi disk is not visible at a depth of about 2-3cm until your tank is full.

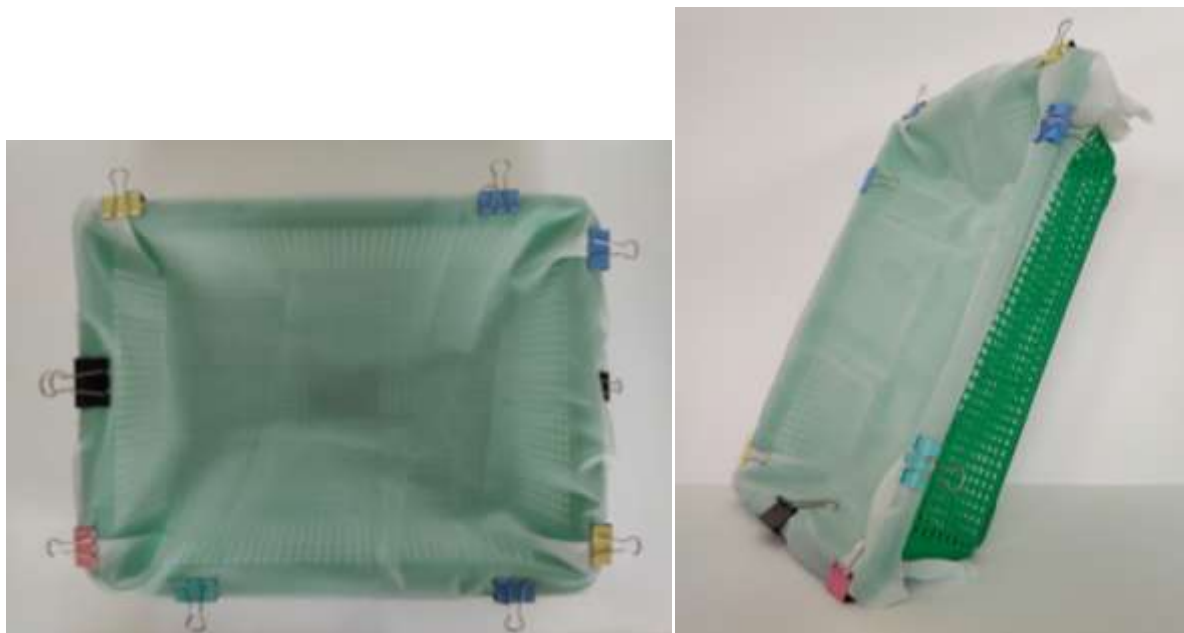
**Important note:** The concentration of spirulina in the culture increases almost linearly with an increase in the turbidity level between the 3.5cm and 1.5cm mark of the Secchi disk. When the Secchi disk is not visible at a depth of 1.5cm, the culture contains spirulina at a concentration of 0.5 g/litre (weight of dried spirulina per litre of culture); when the Secchi disk is visible up to at a depth of 3.5 cm the spirulina is present at a concentration of roughly 0.25 g/liter (Antenna Technologies, 1999).

### 3.3.4 Harvesting and maintaining your culture

1. Pour the volume of the spirulina culture that you would like to harvest into a container. If your cultivation tank has a dispenser valve, this process will be much easier. I personally recommend to harvest about 37% of the culture volume every time its turbidity does not allow to see the Secchi disk below its 2cm mark. This harvesting volume will allow your culture to maintain a healthy turbidity upon adding later on the same amount of fresh culture medium that was harvested to increase again its production. With this harvesting ratio the culture should be able to harvest again in 1 week approximately.

2. Pour the harvested culture over a 30 microns filtration mesh. This will allow you to separate the culture medium (filtrate) from your spirulina.

**Important note:** In order to facilitate the filtration process, I recommend you to set the filtration cloth onto a porous tray (Fig. 9). This will make your filtration process easy and fast.



**Figure9.** Recommended set up for the filtration of home spirulina cultured i.e. a 30 microns filtration mesh attached to a porous tray by clumps.

3. Pour over drinking water over spirulina while it is still on the filter. This will remove the remnants of the alkaline culture medium that are covering your spirulina.

4. At this point, you can enjoy your spirulina directly or freeze it for later use. If you decide to freeze it, I recommend you use ice-cube moulds (**Fig.10a**) in order to prepare individual doses. To this end, add around 2 teaspoons of fresh spirulina into each of the ice-cube moulds. Then, add drinking into each socket containing spirulina until they are filled. At this point, your spirulina will be ready to be frozen for later use (**Fig.10b**).



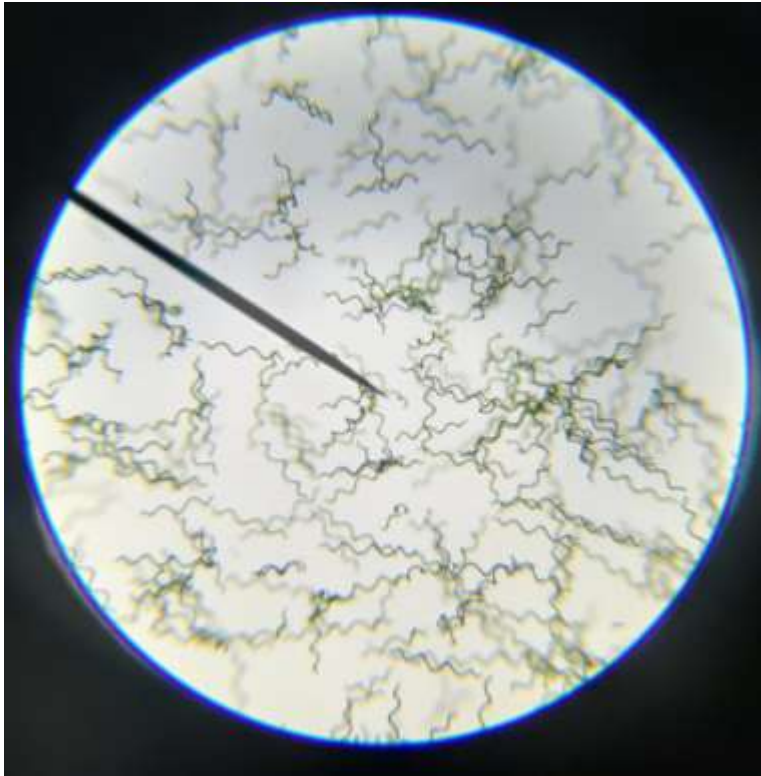
a



b

**Figure10.** Ice-cube moulds for the freezing and storage of spirulina (a). Prepared frozen spirulina cubes for their later use (b).

**Important note:** It is advisable to check under a light microscope a sample of your culture to make sure it has not been contaminated by any invading microorganism. In this respect, the culture should just present spirulina spirals (**Fig. 11**).



**Figure11.**Spirulina culture with no apparent contamination as seen under the light microscope.

5. After harvesting, it is important that you pour new nutrient medium into the cultivation tank so you can keep producing spirulina. To prepare the culture medium, refer to the section **2.3 Nutrients**. It is important that after pouring new culture medium the Secchi disk is not visible at a depth higher than 5cm. This should be easily achieved by adding the same volume of fresh culture medium as the volume of harvested culture if you follow the recommendations mentioned in the point 1 of this section.

**Important note:** If you think that your cultivation tank needs to be cleaned due to the sedimentation of spirulina in certain areas, feel free to do so. In this case, remember to keep your spirulina in a thoroughly disinfected vessel meanwhile you clean your cultivation tank and the rest of the equipment.

## 4. References

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